

Compile-Time Polymorphism in Java (Method Overloading)



Compile-time polymorphism happens **when multiple methods have the same name but different parameters** — either in number, type, or order.

The word "compile-time" means that the Java compiler decides which method to call **based on the method signature**.

Here's the official definition:

Method Overloading in Java means defining **multiple methods with the same name** in the same class, but each method has a **different parameter list**.

CoffeeMachine

makeCoffee()
makeCoffee(String sugar)
makeCoffee(String sugar, milk)

```
// Call each overloaded method
machine.makeCoffee();           // Plain coffee
machine.makeCoffee("sugar");    // Coffee with sugar
machine.makeCoffee("sugar", "milk"); // Coffee with sugar and milk
```

```
// CoffeeMachine class demonstrating method overloading
class CoffeeMachine {

    // Method 1: No parameters
    public void makeCoffee() {
        System.out.println("Making plain coffee...");
    }

    // Method 2: One String parameter
    public void makeCoffee(String sugar) {
        System.out.println("Making coffee with " + sugar + "...");
    }

    // Method 3: Two String parameters
    public void makeCoffee(String sugar, String milk) {
        System.out.println("Making coffee with " + sugar + " and " + milk + "...");
    }
}

// Main class
public class Main {
    public static void main(String[] args) {
        // Create object of CoffeeMachine
        CoffeeMachine machine = new CoffeeMachine();

        // Call each overloaded method
        machine.makeCoffee();           // Plain coffee
        machine.makeCoffee("sugar");    // Coffee with sugar
        machine.makeCoffee("sugar", "milk"); // Coffee with sugar and milk
    }
}
```

```
public class Calculator {

    // Adds two integers
    public int add(int a, int b) {
        return a + b;
    }

    // Adds three integers
    public int add(int a, int b, int c) {
        return a + b + c;
    }

    // Adds two double values
    public double add(double a, double b) {
        return a + b;
    }

    // Main method to test overloaded methods
    public static void main(String[] args) {
        Calculator calc = new Calculator();

        // Call the add method with different arguments
        int sum1 = calc.add(10, 20);           // two integers
        int sum2 = calc.add(5, 15, 25);        // three integers
        double sum3 = calc.add(3.5, 2.7);      // two doubles

        // Print results
        System.out.println("Sum of two integers: " + sum1);
        System.out.println("Sum of three integers: " + sum2);
        System.out.println("Sum of two doubles: " + sum3);
    }
}
```